Evolution, not revolution

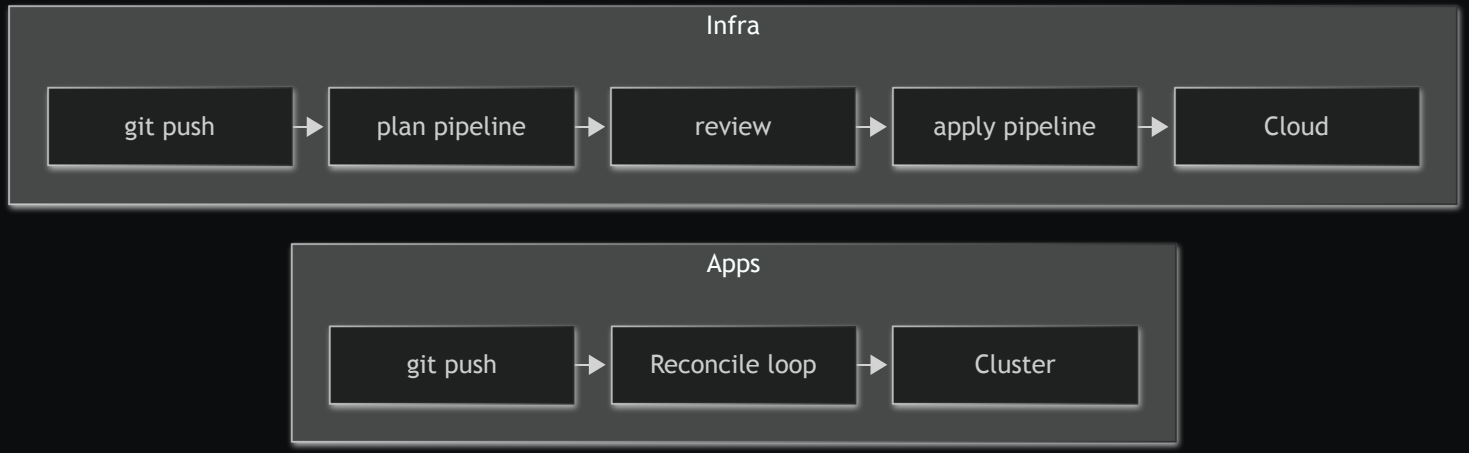
Pushing Terraform into Kubernetes GitOps

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ContainerDay 2026

Your apps: `git push` → **reconciled**

Your infra: `git push` → **plan pipeline** → **manual review** → **apply pipeline**

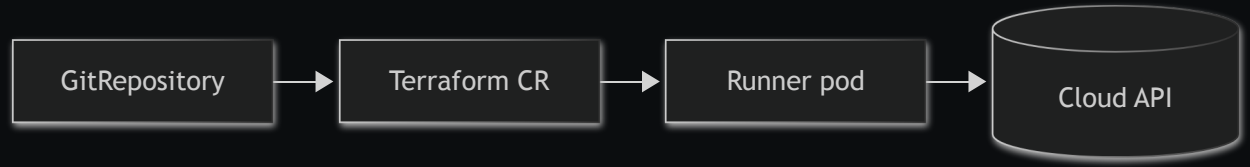


Not a missing automation. **Missing a shared one**

♻️ Don’t rewrite. Run it in Kubernetes

tofu-controller

A Flux controller that runs your Terraform



Introducing Terraform Custom Resource

Point your existing stack, tofu-controller applies it.

```yaml
apiVersion: infra.contrib.fluxcd.io/v1alpha2
kind: Terraform
metadata:
  name: rds
spec:
  path: ./root-modules/rds
  sourceRef:
    kind: GitRepository
    name: infra
  vars:
    - name: size
      value: small
  interval: 10m
  approvePlan: auto
```

🔐 Outputs become Secrets

From state to Secret to Pod

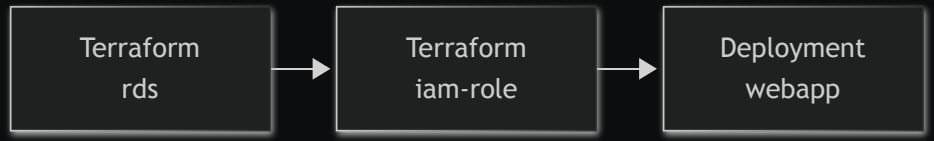
Terraform CR

```yaml
spec:
  writeOutputsToSecret:
    name: rds-output
    outputs:
      - endpoint
      - password
      - db_arn
```

Deployment

```yaml
spec:
  containers:
    - name: webapp
      env:
        - name: DB_ENDPOINT
          valueFrom:
            secretKeyRef:
              name: rds-output
              key: endpoint
        - name: DB_PASSWORD
          valueFrom:
            secretKeyRef:
              name: rds-output
              key: password
```

One dependency graph



`dependsOn`, both ways

Terraform after Terraform

```yaml
apiVersion: infra.contrib.fluxcd.io/v1alpha2
kind: Terraform
metadata:
  name: iam-role
spec:
  dependsOn:
    - name: rds
  varsFrom:
    - kind: Secret
      name: rds-output
      varsKeys:
        - db_arn
```

App after infra

```yaml
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: webapp
spec:
  dependsOn:
    - name: infra
  wait: true
```

🚦 Where did the approval gate go?

Merge, plan, review, approve

```yaml
spec:
  storeReadablePlan: human
  approvePlan: ""
```

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: tfplan-default-rds
  annotations:
    savedPlan: plan-main-b8e362c206
data:
  tfplan: |
    # aws_db_instance.rds will be created
    + resource "aws_db_instance" "rds" {
        + identifier = "rds"
        ...
      }
```

```yaml
spec:
  storeReadablePlan: human
  # plan-<branch>-<10 char git sha>, tied to the exact commit planned
  approvePlan: plan-main-b8e362c206
```

Humans love to click a console

What’s on Git wins anyway

⚠️ Where this gets harder

- Terraform state still required
- Cluster holds infrastructure credentials
- Debugging means watching pod logs
- Drift detection is costly for big stacks
- Still an alpha API, community maintained

One repo, one loop

One control plane

Thanks 🐟

github.com/gionn/tofu-controller-lightning-talk

flux-iac.github.io/tofu-controller

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